Predicting House Prices in Bangalore

Project 1: Regression | Major Project, DSBA September 2026 Batch

Name: Krishna Sanjeev Yadav

1. Objective

Build a regression model that predicts house prices in Bengaluru from area, location and number of rooms, compare several algorithms, and explain the results in business terms.

2. Dataset

The Bangalore House Price Prediction dataset (Kaggle) has 13,320 listings and 9 columns: area_type, availability, location, size, society, total_sqft, bath, balcony and price. Prices are in lakhs of rupees (1 lakh = 100,000 rupees). The data is about 5 years old, so absolute prices are not current.

3. Methodology

3.1 Data cleaning

- Removed 529 duplicate rows (13,320 to 12,791).
- Dropped `society` (41.31% missing).
- Dropped 17 rows with missing `location` or `size`.
- Filled missing `bath` and `balcony` with their medians (both 2.0).
- Created a numeric `bhk` column from `size`.
- Converted `total_sqft` from text to numbers (ranges such as "1133 - 1384" became their average) and dropped 46 rows with unusable units.
- Result: 12,728 rows.

3.2 Outlier removal

Rule	Rows removed
Area per bedroom below 300 sqft	734
Bathrooms more than BHK + 2	9

Rule	Rows removed
Price per sqft outside 1 standard deviation within its location	2,342
Larger BHK cheaper per sqft than a smaller BHK in the same location	1,666
Homes above 10,000 sqft (data-entry errors)	13
Homes with more than 10 bedrooms	4

The dataset went from 12,728 to 7,960 rows (about 37% removed). This gives cleaner data at the cost of fewer rows, so the model is intended for typical homes, not extreme listings.

3.3 Feature engineering

- Created `price_per_sqft` for outlier detection and EDA only. It is calculated from the target, so it was **not** used as a model input (this would be data leakage).
- Grouped locations with 10 or fewer listings into "other": 1,193 locations became 170.
- Simplified `availability` from 81 values to Ready To Move / Not Ready.
- Dropped `size` (replaced by `bhk`).
- Price is heavily right-skewed (skew 7.06), so a log-price version was also tested.

3.4 Exploratory analysis (14 plots plus a bonus)

- `total_sqft` has the strongest correlation with price (0.78), followed by `bath` (0.55) and `bhk` (0.51). `balcony` is weak (0.15).
- `bhk` and `bath` are highly correlated (0.87).
- Median price rises up to 4 BHK and then levels off, so the relationship is not a straight line.
- Location matters: central areas such as Rajaji Nagar, Benson Town and Indira Nagar average about 12,000 to 14,600 rupees per sqft, while the cheapest areas average about 2,900 to 3,800.
- Availability has almost no effect on price (median about 70 lakhs for both groups).

3.5 Modelling

The data was split 80/20 (random_state = 42) before any scaling or encoding. A scikit-learn pipeline scaled the numeric features and one-hot encoded location, area type and availability, so the test set never influenced preprocessing. Models were scored with RMSE, MAE and R^2 on the test set, plus 5-fold cross-validation R^2 on the training set.

4. Results

Model	Train R^2	Test R^2	RMSE	MAE	CV R^2
Random Forest (tuned)	0.856	0.770	47.58	22.13	0.694
Random Forest	0.938	0.734	51.18	21.80	0.672
Gradient Boosting	0.812	0.734	51.15	23.85	0.671
Ridge	0.659	0.690	55.22	27.57	0.658
Linear Regression	0.659	0.689	55.29	27.66	0.658
Lasso	0.650	0.684	55.69	28.18	0.651
Decision Tree	0.983	0.387	77.62	28.00	0.542

RMSE and MAE are in lakhs of rupees.

Key observations

- The Decision Tree overfits badly (train R^2 0.983 vs test R^2 0.387).
- Tree ensembles beat linear models by about 0.045 R^2, consistent with the non-linear pattern seen in EDA.
- Ridge matched Linear Regression, so the 0.87 BHK/bath correlation did not hurt much.
- Log-transforming price did not help (Random Forest test R^2 0.717 vs 0.734). Linear Regression on log price failed (CV R^2 -0.699), most likely because exponentiating linear predictions amplified a few large errors. Raw price was kept.
- Hyperparameter tuning (RandomizedSearchCV, 12 combinations, 3-fold CV on training data) raised test R^2 from 0.734 to 0.770, cut RMSE by about 7%, and shrank the train/test gap from 0.204 to 0.086. Best settings: n_estimators = 200, max_depth = 15, min_samples_split = 2, min_samples_leaf = 2, max_features = 0.5.
- Cross-validation R^2 (0.694) is below test R^2 (0.770), so it is the more conservative estimate of real performance.

5. Interpretation and business insights

Top 5 features (permutation importance): total_sqft (by far the strongest), location, area_type, bath, bhk. balcony and availability add almost nothing.

- **Size:** in Whitefield, a 2 BHK is predicted at 50.7 lakhs for 1,000 sqft and 83.6 lakhs for 1,400 sqft, roughly 8 lakhs per extra 100 sqft.
- **Location:** the same 1,200 sqft, 2 BHK home is predicted at 120.5 lakhs in Rajaji Nagar, 77.1 in Indira Nagar, 70.2 in Whitefield and 63.2 in Electronic City, about 1.9 times higher in Rajaji Nagar than in Electronic City.

- **For buyers:** a fixed budget buys noticeably more area in outer areas than in central ones.
- **For builders and agents:** price mainly by area first and location second. Balcony count and availability barely move price in this data.
- **Accuracy:** the typical error is about 22 lakhs, so the model suits quick estimates for typical homes.

6. Limitations

- The data is about 5 years old and contains listing prices, not final sale prices.
- Floor, building age, amenities and distance to metro or IT parks are missing.
- About 37% of rows were removed as outliers, so the model is not suited to extreme listings.
- Errors grow with price. Homes above about 500 lakhs are consistently under-predicted (for example, an actual 1,500 lakhs predicted at about 980) because luxury homes are rare in the training data.
- Premium areas look under-predicted compared with their average price per sqft, since tree models pull predictions toward the average.
- Locations with 10 or fewer listings were grouped as "other".

7. Conclusion

The tuned Random Forest was the best model (test R^2 0.770, RMSE 47.6 lakhs, MAE 22.1 lakhs, CV R^2 0.694). It captures the non-linear link between size, rooms and price better than linear models, and it avoids the heavy overfitting of a single Decision Tree. Area is the dominant price driver, followed by location. Future work could add building age, floor and distance-to-metro features, use more recent data, and model luxury homes separately.